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The cyclonic circulation in the Australian–Antarctic basin simulated by an eddy-resolving general circulation model

Shigeru Aoki · Yoshikazu Sasai · Hideharu Sasaki · Humio Mitsudera · Guy D. Williams

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Abstract Flow structure in the Australian–Antarctic basin is investigated using an eddy-resolving general ocean circulation model and validated with iceberg and middepth float trajectories. A cyclonic circulation system between the Antarctic Circumpolar Current and Antarctic Slope Current consists of a large-scale gyre in the west (80–110° E) and a series of eddies in the east (120–150° E). The western gyre has an annual mean westward transport of 22 Sv in the southern limb. Extending west through the Princess Elizabeth Trough, 5 Sv of the gyre recirculates off Prydz Bay and joins the western boundary current off the Kerguelen Plateau. Iceberg trajectories from QuickScat and ERS-1/2 support this recirculation and the overall structure of the Antarctic Slope Current against isobath in the model. Argo float trajectories also reveal a consistent structure of the deep westward slope current. This study indicates the presence of a large cyclonic circulation in this basin, which is comparable to the Weddell and Ross gyres.

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S. Aoki (✉) · H. Mitsudera · G. D. Williams
Institute of Low Temperature Science, Hokkaido University, N19W8, Kita-ku, Sapporo 060-0819, Japan
e-mail: shigeru@lowtem.hokudai.ac.jp

Y. Sasai
Frontier Research Center for Global Change, Japan Agency for Marine-Earth Science and Technology, Yokohama, Japan

H. Sasaki
Earth Simulator Center, Japan Agency for Marine-Earth Science and Technology, Yokohama, Japan

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1 Introduction

The Antarctic Divergence plays a central role in the meridional overturning circulation in the Southern Ocean. Bounded by circumpolar westerlies centered near 50° S and polar easterlies near the Antarctic coast, a trough of surface atmospheric pressure surrounds Antarctica. The surface Ekman transport, imposed by the wind system, is divergent in this latitude band and causes the uplift of warm/saline deep water that partially originated in the North Atlantic. The wind system and basin configuration result in the subpolar gyre system of the Weddell and Ross Seas (e.g., Rintoul et al. 2001), bounded by the Antarctic Circumpolar Current (ACC; e.g., Whitworth and Nowlin 1987; Orsi et al. 1995) to the north and the Antarctic Slope Current (ASC; e.g., Jacobs 1991; Whitworth et al. 1998) to the south. The main Weddell Gyre extends zonally from the continental slope off the Antarctic Peninsula to 30° E (Deacon 1979; Gouretski and Danilov 1993) with an outer limb extending as far as 60° E (Park and Gamberoni 1995). It has a transport of about 50 Sv on the Greenwich Meridian (Klatt et al. 2005). The Ross Gyre extends from 170° E to 140° W (Gouretski 1999), although its volume transport is not well defined compared to that of the Weddell Gyre.

In the Indian Ocean sector, the negative wind stress curl associated with the Antarctic Divergence is also present and indicative of a cyclonic oceanic substructure. Zeverev (1963) observed the presence of a

series of cyclonic gyres between 20° and 100° E. Smith et al. (1984) showed that subsurface cyclonic eddies of 500 km diameter were located at 67° and 80° E along 65° S. Wakatsuchi et al. (1994) revealed the presence of cyclonic eddies at 67°, 100°, and 115° E, and the zonal diameter of the eddies was similarly estimated to be approximately 500 km. Using satellite images and higher-resolution hydrographic data, the presence of smaller cyclonic eddies with 130 km diameters was revealed in the Antarctic Divergence region off the Adélie Land Coast in 130–145° E (Fig. 1; Hirawake et al. 2003; Aoki et al. 2007). However, the general structure of these cyclonic features is not clearly understood on a basin scale.

Increasing attention has been given to the presence of a larger "gyre" structure in the Australian–Antarctic basin, equivalent to the Weddell and Ross gyres. Park and Gamberoni (1995) presented the western end of a cyclonic surface circulation in this basin using satellite altimetry and an ocean general circulation model (OGCM). Using several hydrographic transects between about 63° S and the coast, Bindoff et al. (2000) revealed more direct evidence of a cyclonic gyre extending from 90° to 110° E and estimated a westward transport of 29 Sv along the slope and rise. More recently, McCartney and Donohue (2007) reported that a deep cyclonic gyre extends between 85° and 150° E, based mainly on two full-depth WOCE transects. They derived a westward total (barotropic + baroclinic) transport of 76 Sv along the coast, half of which is transported along the slope and the other half recirculates. This estimate prescribes an average current of 30 cm/s over the continental slope. This is much larger than that obtained by Bindoff et al. (2000) and the deep current estimate of 5–16 cm/s derived from a year-long mooring of current meters on the continental slope off the Adélie Land Coast (Fukamachi et al. 2000). Hence, a further examination of the circulation structure and its transport in this region is necessary.

Some OGCMs have reproduced a similar large-scale cyclonic circulation in the Australian–Antarctic basin. From the BRIOS-1 model (Beckmann et al. 1999), the annual mean stream function revealed a cyclonic "Kerguelen Gyre" of more than 30 Sv in the western half of the basin (Rodehacke et al. 2007); this is a part of the closed cyclonic contour extending from the Weddell Gyre. The transport of the gyre was larger than that of the Ross Gyre simulated by the same model. Using the results of coarse-resolution Intergovernmental Panel on Climate Change (IPCC) Fourth Assessment Report coupled general circulation models, Wang and Meredith (2008) revealed that the cyclonic gyre in this basin differs considerably among the simulations; transport ranges from 0 to 32 Sv, with a few models showing a presence of the western boundary current. The barotropic stream function of the eddy-permitting OCCAM model (Webb et al. 1998) showed a northward bend (hence western boundary current) off the Kerguelen Plateau and southward migration eastward (Plate 4.6.6 in Rintoul et al. 2001). A triangle-shaped cyclonic circulation of more than 15 Sv was found from 90° to 130° and/or 145° E. Lagrangean tracer analysis of this model revealed a cyclonic circulation centered at 120° E (Thorpe et al. 2007). Near the southern

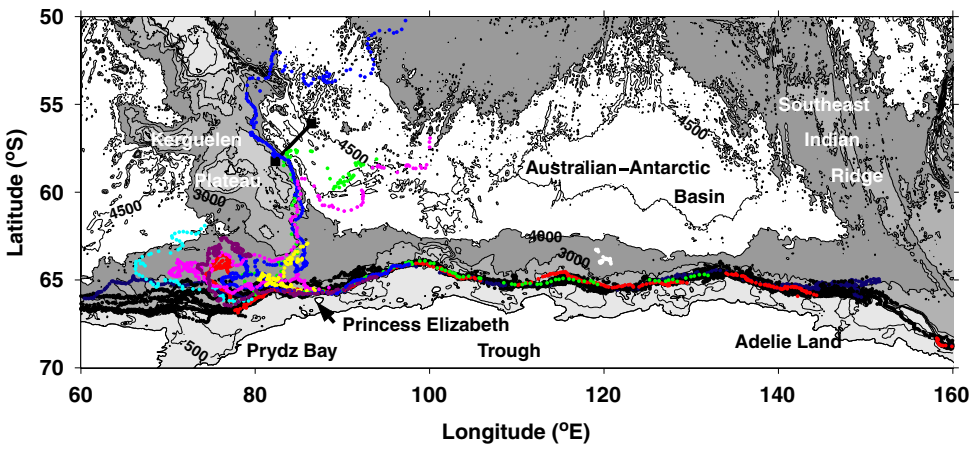

Fig. 1 Iceberg trajectories (*black dots*) around East Antarctica from the Brigham Young University database (total 34) with ETOPO2 bathymetry. *Colored data points* denote specific icebergs; B-17B (*red*), C-6 (*green*), C-19C (*dark blue*), C-20 (*blue*), E-15 (*cyan*), UK-95 (*magenta*), UK-110 (*yellow*), UK-116 (*purple*), and UK-242 (*white*). Oceanic regions shallower than 4,000 m are *shaded*. The *thick line between two black squares* near 83° E, 57° S indicates the hydrographic transect in 2005 off the Kerguelen Plateau used in this study

boundary, there was westward flow along the slope and recirculation within the Princess Elizabeth Trough (PET; Fig. 1). Rintoul et al. (2008) used this current along the Kerguelen Plateau to explain anomalous sea ice drift south of the Kerguelen Plateau. While the presence of a cyclonic gyre is somewhat common in these numerical models, the detailed vertical profile and horizontal structure of the current and transport are not thoroughly investigated, especially near the continental slope. Moreover, quantitative comparisons with observations, in particular with respect to seasonal variability, are still lacking due to the undersampling caused by the seasonal sea ice cover.

To investigate the cyclonic circulation in the Australian–Antarctic basin, we examine the output of the ocean general circulation model for the earth simulator (OFES), which is a high-resolution OGCM (Masumoto et al. 2004). It has successfully simulated the major current systems and associated mesoscale eddies such as the Kuroshio, Kuroshio Extension, and Agulhas current. The annual mean transport of the ACC across the Drake Passage is about 150 Sv, which is only slightly larger than the overall estimate of 134 ± 13 (Whitworth 1983; Whitworth and Peterson 1985; Rintoul et al. 2001). In the Australian–Antarctic basin, the generation of cyclonic eddies off the Adélie Land coast was detected, and it compared well with in situ hydrographic and satellite observations (Aoki et al. 2007). As such, it is potentially a useful tool for capturing realistic flow structure over the whole basin.

To evaluate the OFES flow structure, we use two Lagrangean-type observational datasets, iceberg, and middepth float trajectories. Historically, iceberg drift trajectories have been used to examine the surface current on the slope and its variation (e.g., Gordon and Tchernia 1972; Tchernia and Jeannin 1980; Ohshima et al. 1996; Gladstone et al. 2001). However, quantitative estimates were difficult due to the uneven sampling of the data. Recently, the deployment of satellite scatterometers to polar regions has made it possible to monitor snow and ice conditions continuously at short intervals and homogeneously around Antarctica (e.g., Long and Drinkwater 1994, 1999; Remund and Long 1999). Aoki (2003) described the spatial pattern and seasonal variation of the ASC in the Weddell–Enderby basin from satellite-tracked iceberg positions and proved their usefulness. For the deeper currents, middepth floats are useful although sea ice prevents operation in winter time in this study. Argo floats were used to describe the vertical structure of the current system.

The aim of this paper is to map the gyre-scale circulation pattern in the Australian–Antarctic basin using the OFES model output, together with iceberg and middepth float trajectories. The circulation system obtained is discussed in connection with the in situ observations and previous studies. It is noted that the westward current along the continental slope is a key component of the current system in this study, but there are some differences in terminology and definitions in literatures, including the terms like the east wind drift and the Antarctic Coastal Current. We use the term ASC for the geostrophic current on the continental slope associated with the Antarctic Slope Front, according to Whitworth et al. (1998), and distinguish it from the broader westward current on the continental slope and rise.

2 Study region and data source

2.1 Study region

The Australian–Antarctic basin is semi-enclosed with the Antarctic continent to the south, the Southeast Indian Ridge to the north and east, and the Kerguelen Plateau to the west (Fig. 1). It can be divided into western and eastern depressions separated at around 100–110° E; a closed isobath of 4,500 m depth exists between 85° and 95° E just east of the Kerguelen Plateau and between 110° and 140° E northwest of the Adélie Land Coast.

The Kerguelen Plateau is a prominent bathymetric feature with islands that can be separated into northern and southern parts. The depth of the shallowest region of the southern plateau is less than 1,000 m. The northern plateau has a shallow rise projected southeastward from the main plateau, which is called the Chun Spur. In the middle of the plateau, there is a col called the Fawn Trough.

Between the Kerguelen Plateau and the continent, there is a trough called the PET. The sill depth is about 3,700 m and the deepest part is located around 64–65° S. Southwest of the PET, there is a large continental embayment called Prydz Bay, which holds the Amery Ice Shelf.

The Southeast Indian Ridge that consists of various topographic features partly acts as the eastern boundary. The southern end of the ridge is the Balleny Islands at 66–67° S, whose northern trough is around 3,000 m deep and the southern trough is around 2,500 m deep.

The continental shelf/slope runs almost zonally in this sector, with slightly northward migrations around 100, 115, and 135° E. The continental slope deepens steeply to around 3,000 m, and the continental rise subsequently deepens toward 58–60° S to 4,000–4,500 m.

2.2 The OFES numerical model and experiment

The OFES numerical model is based on the Geophysical Fluid Dynamics Laboratory's Modular Ocean Model (Pacanowski and Griffies 2000). The model domain is from 75° S to 75° N with horizontal grid spacing of 1/10°. The model topography is constructed from the 1/30°bathymetry dataset created by the OCCAM project. A detailed description of the model formulation is given by Masumoto et al. (2004). The results used in this study are those of the 47th year of the 50-year integration, forced with the climatological monthly mean wind stress, heat flux, and fresh water flux from the National Centers for Environmental Prediction (NCEP)/National Center for Atmospheric Research reanalysis data. Annual and monthly mean flow fields were analyzed because the interannual variability is relatively small. Sea ice processes are not explicitly included in this model. However, we adopted restoring to the monthly mean sea surface salinity of the World Ocean Atlas 1998 (e.g., Boyer et al. 1998) with a restoring time scale of 6 days to include the contributions from the sea ice and land ice/water.

2.3 Iceberg trajectories

Iceberg drift is a useful proxy for inferring the surface current around Antarctica. Trajectories of icebergs were derived from satellite scatterometer images with positions from a database at Brigham Young University (http://www.scp.byu.edu/data/iceberg/database1.html). The positions of 34 icebergs from the database were used (Table 1). The icebergs were first detected at various regions along the coast; 17 icebergs started east of 140° E, 14 icebergs from 90° to 140° E, and three icebergs started west of 90° E. The data obtained by

Table 1 Drift information for the 34 icebergs taken from the Brigham Young University database: start and end time and positions of each iceberg

Code	Start date	Position	End date	Position
B-9A	25 January 1992	133.3° E, 65.4° S	28 December 1999	56.0° E , 65.6° S
B-15B	4 May 2000	169.9° E, 78.2° S	3 August 2008	77.5° E, 67.5° S
B-15D	18 August 2000	179.0° E, 76.4° S	3 August 2008	35.0° E, 56.4° S
B-15F	12 December 2000	179.4° E, 75.0° S	3 August 2008	55.1° E, 65.6° S
B-15G	4 May 2001	170.1° E, 71.0° S	3 August 2008	48.3° E, 66.4° S
B-15I	8 October 2001	157.3° E, 68.5° S	21 December 2002	122.7° E, 65.6° S
B-15K	19 February 2004	166.8° E, 76.5° S	3 August 2008	69.5° E, 66.2° S
B-15N	1 November 2005	171.6° E, 71.6° S	3 August 2008	78.9° E, 66.5° S
B-16	25 April 2001	152.1° E, 68.2° S	19 December 2007	131.5° E, 65.1° S
B-17A	26 December 2000	178.5° E, 73.9° S	3 August 2008	127.1° E, 65.3° S
B-17B	25 December 2000	179.9° E, 73.5° S	3 August 2008	77.1° E, 64.7° S
C-6	15 February 1992	131.9° E, 64.7° S	15 November 1993	93.9° E, 58.1° S
C-8	21 July 1999	147.1° E, 67.3° S	3 August 2008	54.0° E, 65.5° S
C-9	21 July 1999	146.5° E, 67.0° S	8 December 2004	50.7° E, 66.0° S
C-10	21 July 1999	98.1° E, 64.4° S	13 July 2006	25.4° E, 70.1° S
C-16	14 September 2000	170.5° E, 77.1° S	3 August 2008	116.7° E, 66.0° S
C-18A	1 June 2005	115.3° E, 65.2° S	16 February 2008	42.2° E, 73.9° S
C-18B	16 October 2003	153.4° E, 67.7° S	3 August 2008	116.7° E, 66.4° S
C-19B	5 December 2002	174.8° E, 74.2° S	31 December 2006	98.4° E, 64.3° S
C-19C	22 June 2003	162.8° E, 69.4° S	3 August 2008	60.4° E, 66.0° S
C-19D	22 June 2003	163.3° E, 69.5° S	3 August 2008	42.2° E, 72.2° S
C-20	16 March 2003	98.7° E, 65.1° S	6 March 2006	106.2° E, 47.7° S
C-12	20 March 2005	90.3° E, 66.1° S	7 July 2008	41.2° E, 54.7° S
E-15	13 April 1998	79.6° E, 66.1° S	9 December 1998	74.2° E, 61.9° S
UK-90	30 June 2002	94.3° E, 64.9° S	12 October 2002	90.4° E, 64.9° S
UK-93	03 August 2003	94.3° E, 64.4° S	3 August 2008	80.9° E, 66.6° S
UK-95	20 July 2003	88.5° E, 65.4° S	12 July 2004	100.0° E, 56.9° S
UK-110	07 August 2004	86.9° E, 65.6° S	19 January 2005	85.9° E, 63.0° S
UK-112	07 July 2004	93.9° E, 64.8° S	6 December 2004	79.0° E, 66.5° S
UK-116	12 July 2004	93.9° E, 64.8° S	18 December 2006	78.9° E, 63.6° S
UK-134	29 June 2005	123.6° E, 65.5° S	8 January 2006	112.4° E, 65.5° S
UK-144	17 February 2006	138.2° E, 65.4° S	16 April 2006	132.4° E, 64.9° S
UK-173	11 January 2005	134.2° E, 65.1° S	3 August 2008	121.5° E, 65.7° S
UK-242	30 March 2008	119.3° E, 63.3° S	13 May 2008	120.8° E, 64.1° S

QuickScat from 20 July 1999 to 2 August 2008 and ERS-1/2 from 25 January 1992 to 2 January 2000 were used.

Icebergs that drifted near the coast frequently became grounded and showed no motion for a long period of time. Hence, drift data at ocean depth shallower than 500 m were excluded in the subsequent analysis. For this screening, the digital database of 2-min Gridded Global Relief Data (ETOPO2) from the National Geophysical Data Center, National Oceanic and Atmospheric Administration, U.S. Department of Commerce was used.

2.4 Middepth float trajectories

Middepth floats are useful tracers that can describe the deep flow field. However, the data coverage is sparse in the regions near the Antarctic margin and east of the Kerguelen Plateau where seasonal sea ice exists. The trajectories of Argo floats were used. The drift data from 53 floats were taken from the Argo project; most of the floats were located in the northern half of the region. Parking depth was set mostly to 1,000 m and the repeat interval was 10 days.

3 Cyclonic circulation in the OGCM

The fine-resolution OGCM can simulate realistic properties of the circulation structure. Here, we examine the spatial structure and seasonal variability of the cyclonic circulation in the Australian–Antarctic basin revealed in the OFES results.

3.1 Surface current system

Annual mean surface elevation shows a zonally asymmetric circulation in the Australian–Antarctic basin, with the ACC to the north and the ASC to the south (Fig. 2). The southern part of the ACC bends sharply equatorward passing through the PET around 85° E, forming a jet along the slope off the Kerguelen Plateau, and recirculates poleward immediately offshore. The southern part of the ACC shifts poleward around 105° E while the overall ACC axis gradually shifts poleward at the same time. The surface current is stagnant in the interior region of 90–105° E. To the south, the ASC is found along the upper continental slope, flowing westward. After penetrating into the PET, some portion returns northeastward to join the western boundary current off the Kerguelen Plateau. Associated with this northeastward flow, a convergence of elevation contours at around 61° S, 87° E is detected.

3.2 ASC, westward current on the slope, and ACC

For each part of the current system, we will examine the quantitative aspects of the vertical/horizontal structures and their seasonal variation. Profiles of zonal velocity show the detailed structure of the current on the slope/rise (Fig. 3). At 100° E, the annual mean profile shows an intense ASC on the upper slope; the most prominent core is surface-intensified (about 30 cm/s) with a width of 50–60 km. Weak westward flow is found on the lower slope throughout the water column. Off the slope, weak westward flow continues to 60.5° S, and the flow direction changes to eastward north of 60.5° S. At 120° E, the annual mean profile shows the intense ASC (~30 cm/s) on the upper slope as well. General westward flow with a bottom-intensified structure is found on the lower slope. An eastward current down to 2,000 m is found near 63–63.5° S, showing a southward shift compared to 100° E. Below this current, westward flow is detected near the bottom north to 62° S. At 140° E, a shallow westward core (20 cm/s) of the ASC

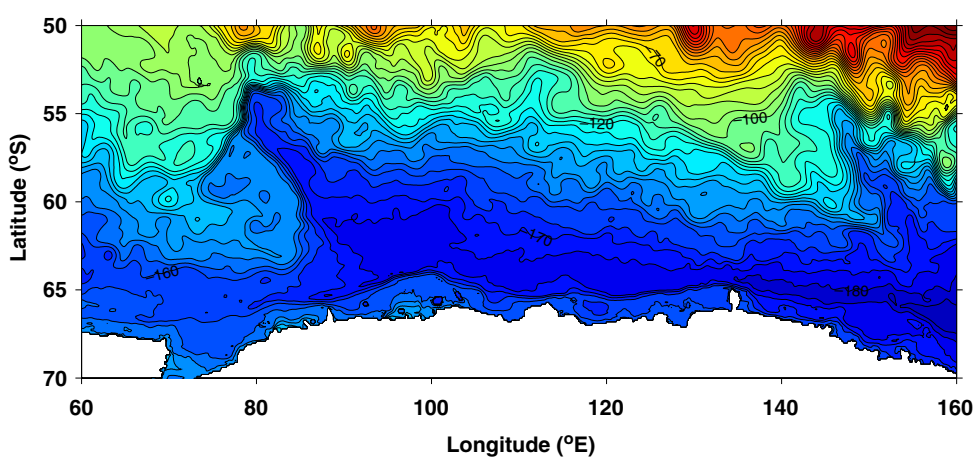

Fig. 2 Annual mean surface elevation (centimeters) of the ocean general circulation model for the earth simulator around East Antarctica. Contour interval is 5 cm

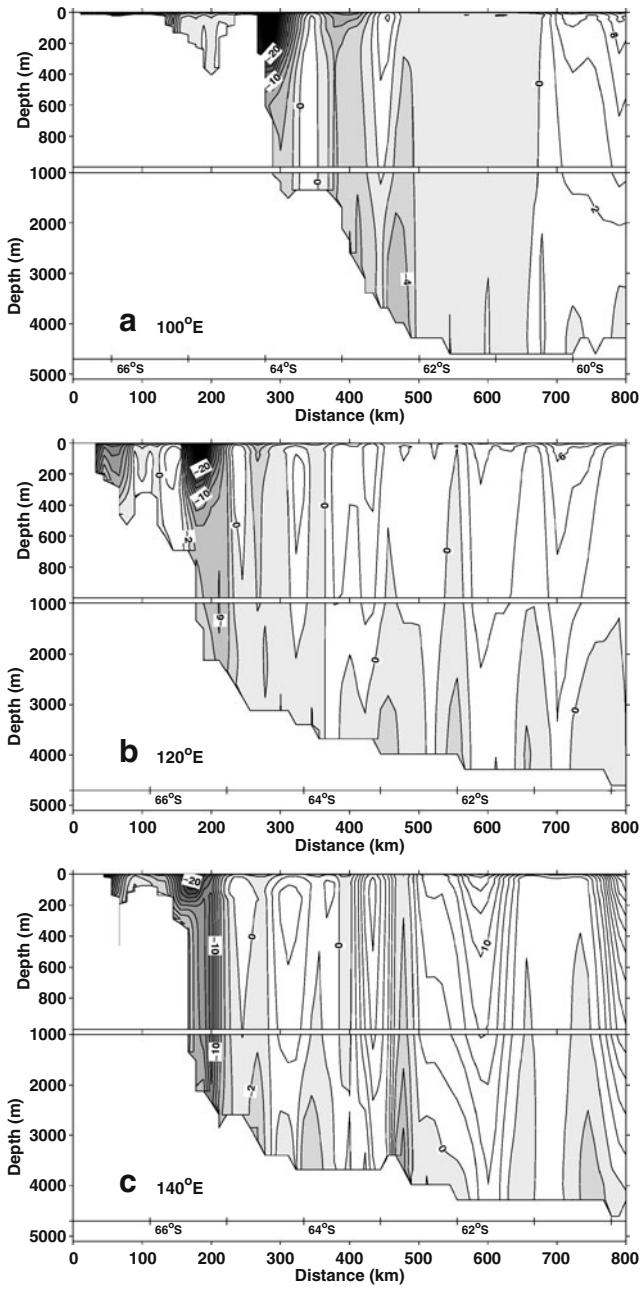

Fig. 3 Vertical sections of the zonal velocity (centimeters per second) across meridional transects at **a** 100° E, **b** 120° E, and **c** 140° E from the OFES. Contour interval is 2 cm/s and negative (westward) values are *shaded*

is found at the shelf break, indicating continuity along the slope. A deep barotropic core (10 cm/s) down to 2,500 m depth is detected. Weak westward current layer of 1,000–1,500 m thickness is obtained near the bottom; this feature is not significant in the western sections. An eastward current, which corresponds to the ACC, is found near 64.5° S, showing its southward shift compared to the west. Near the bottom, westward flow is found over the end of the slope/rise.

To derive the zonal mean property of the current on the slope, zonal velocities near the surface, at a deeper depth (1,500 m), at the bottom, and of depth average are averaged on isobaths for three subregions between 90–110°, 110–130°, and 130–145° E. The origin of the cross-front coordinate is set to where the continental slope reaches 1,500 m depth from the south for the first time, and the distance of each depth was calculated meridionally from that point.

For 90–110° E, the ASC in the surface layer (averaged from surface to 260 m) is around 15 cm/s and centered at 1,500 m depth (Fig. 4a). The width of the mean core is 100–150 km. The deep current is weaker (about 3 cm/s) on the slope. The bottom current is westward and its magnitude is about 2 cm/s. On the lower slope 50–100 km north of the main axis, the current is relatively barotropic. The mean deep current is about 1/3 of the magnitude of the strong surface core at the main axis of the ASC. In 110–130° E, the structure is similar, with a slightly weaker surface core and stronger deep current (Fig. 4b). For 130–145° E, the surface core is similar (~15 cm/s) to that in the west (Fig. 4c). However, the deep current is significantly larger (10 cm/s) at 0–50 km north of the axis and is very barotropic. About 150 km north of the current axis, the current direction is mostly eastward although a weak westward flow is present at the bottom.

A strong seasonal variation of the ASC is detected in the model. Between 90–110° E, the surface current velocity is the largest in winter; it is 22 cm/s in July–September while 11 cm/s in January–February (Fig. 5a). The main axis moves 20 km inshore in association with this intensification. The winter increase in velocity is significant on the upper slope region while the magnitude of the increase gradually decreases toward the lower slope region. For 110–130° and 130–145° E, similar seasonal changes are obtained for the ASC core: maximum in autumn/winter (around 20 cm/s) and minimum in summer (around 10 cm/s; Fig. 5b, c).

3.3 Current structure in the western boundary region

As the Kerguelen Plateau and PET act as a partial barrier, the circulation in the western boundary is complicated. Distribution of the surface elevation shows a penetration of low height inferring a recirculation through the PET (Fig. 2).

To examine the subsurface current structure of this recirculation, horizontal distributions of current vectors are shown at 207, 527, 1,041, and 3,114 m depths. In the northern part, subsurface velocity fields down to

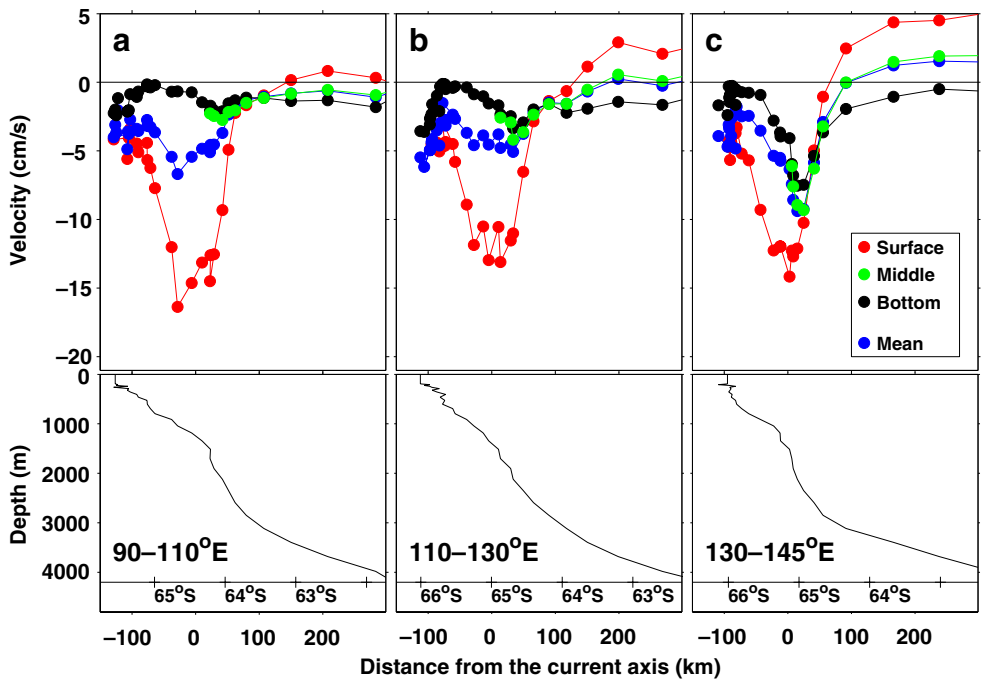

Fig. 4 Spatial variation of the Antarctic Slope Current in the OFES. Data presented as the meridional variation in vertical structure of the mean zonal velocity (centimeters per second) over the continental slope in regions **a** 90–110° E, **b** 110–130° E, and **c** 130–145° E. The abscissa is presented as latitude and meridional distance from the 1,515-m isobath. Velocities presented for the near-surface layer between 5 and 262 m (*red lines*), at 1,515 m (*green*), the *bottom* (*black*), and the total depth-average (*blue*) for the depth average. The *lower panels* show the averaged isobath shape within the regions

1,000 m show that part of the eastward ACC flows south of the Kerguelen Plateau with two prominent fronts along 64° and 63° S (Fig. 6). The currents meander in response to a cyclonic feature at 65° S, 76° E and an anticyclonic feature at 64.5° S, 82° E. Strong eastward flow is still detectable at 3,000 m depth. These fronts bend equatorward around 84° E along the eastern flank of the plateau. In the southern part, the ASC is prominent at 200 and 500 m depths along the slope (66–67° S). An along-coast current is revealed within Prydz Bay. At 200 m depth, it flows southward at 78° E, then westward along the ice shelf/coast edge, and finally northward at 72° E. At 500 m depth, a cyclonic circulation within the deeper trough of the bay is revealed, although it is weaker. In the western end around 68° E, a prominent northward current is found at 200 and 500 m depths, which is also partly revealed in the surface elevation data.

To reveal the vertical structure of these components, profiles of the currents are examined near the slope in the PET along 80° E and along a transect off the Kerguelen Plateau from 82.3° E, 58.2° S to 86.4° E, 56.1° S (Fig. 7; the location of the latter transect is indicated in Fig. 1). The westward current is detected on the

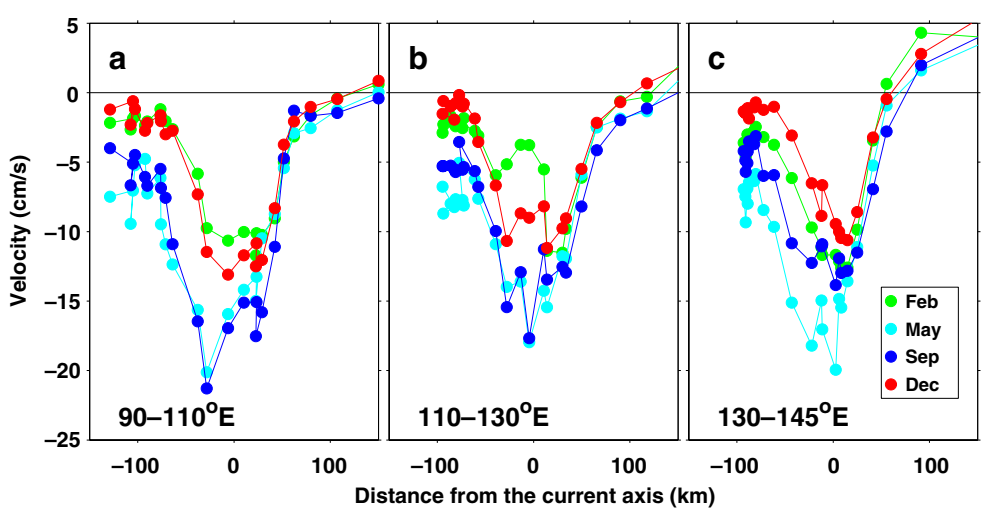

Fig. 5 Seasonal variation of the Antarctic Slope Current in the OFES near the surface in regions **a** 90–110° E, **b** 110–130° E, and **c** 130–145° E. The abscissa is presented as meridional distance (kilometers) from the 1,515-m isobath. Velocity data are shown for February (*green lines*), May (*cyan*), September (*blue*), and December (*red*)

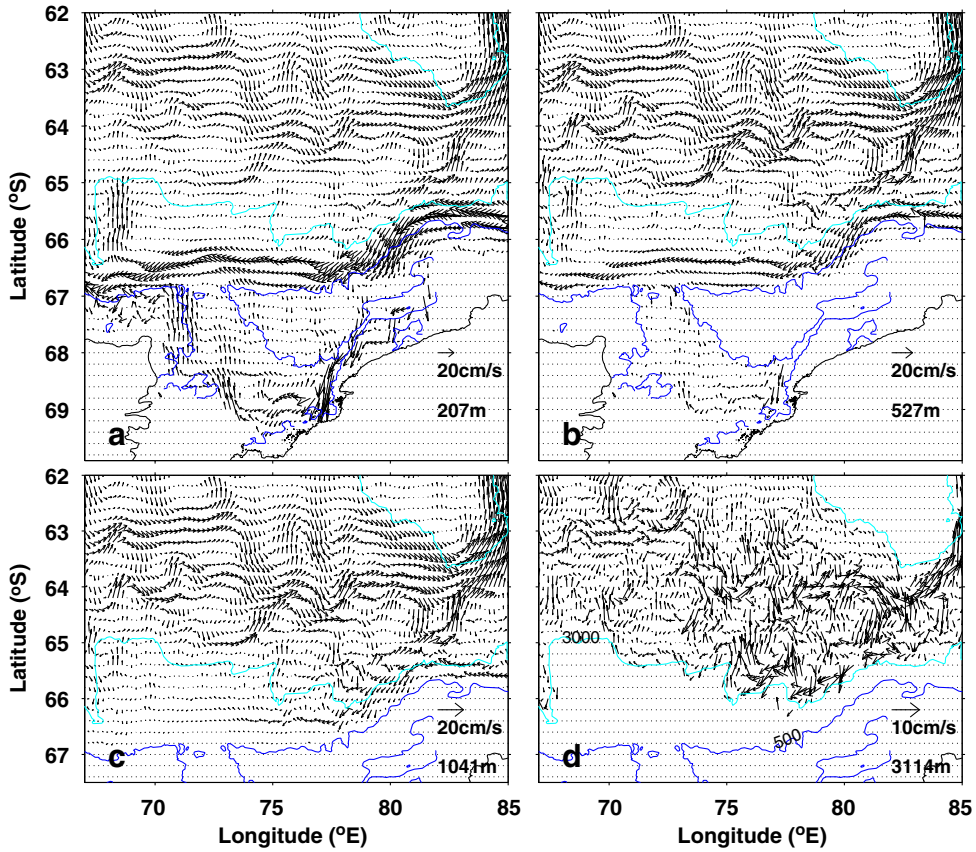

Fig. 6 Ocean circulation near Prydz Bay from the OFES. Vectors of horizontal velocities (centimeters per second, scale as indicated) at **a** 207 m, **b** 527 m, **c** 1,041 m, and **d** 3,114 m depths

slope south of 65.5° S. A strong ASC near the surface is found on the upper slope, while a bottom-intensified westward current is detected on the lower slope (3,000–3,500 m depths). The current direction changes to eastward north of 64.5° S with another eastward core at 65–65.5° S. At the end of the slope between 64.5°and 65° S, the westward velocity is stronger near the bottom compared to that near the surface. Along the Kerguelen Plateau, an equatorward current of 20–30 cm/s is found from the surface down to 1,500 m. Its counter current of 15 cm/s is detected down to 1,500 m from the slope to about 200 km offshore, which carries the ACC transport.

3.4 Depth-integrated transport

The depth-integrated transport is calculated for the Australian–Antarctic basin (Fig. 8). A prominent cyclonic pattern extends from 85° to 110° E. The transport of the cyclonic circulation was 22 Sv, integrating along 100° E from the coast to its center at 60.5° S.

In the east of this circulation, the stream line shifted southward and continued further to 150° E although the eastern closure cannot be clearly defined. The westward transport integrated from the coast to 64.5° S along 140° E is 14 Sv. Between 130° and 150° E, the circulation is not like a large-scale gyre but like a series of mesoscale disturbances.

In the western boundary region, transport consisted of the westward transport along the slope, eastward transport near the Kerguelen Plateau, and a recirculation in between. Along 84° E, westward transport from the coast to 65° S was 10 Sv and eastward transport from 65° to 62.5° S was 23 Sv. Hence, the net transport in the PET was 13 Sv eastward at this longitude. At 65° E, westward transport of 4 Sv continues along the slope, and a remaining 5 Sv recirculates to the east; the recirculation through the PET carries about 5 Sv. Transport within Prydz Bay is not large compared to that of the ASC. The recirculation joins the ACC transport eastward. The transport along the transect off the Kerguelen Plateau is 33 Sv, combining 23 Sv of the ACC transport from the west and 10 Sv within the Australian–Antarctic basin.

Seasonal variation is examined for the depth-integrated transport (Fig. 9). Strong seasonal variability is revealed for the southern limb. In the west, the westward transport increases to 35 Sv in July, and it

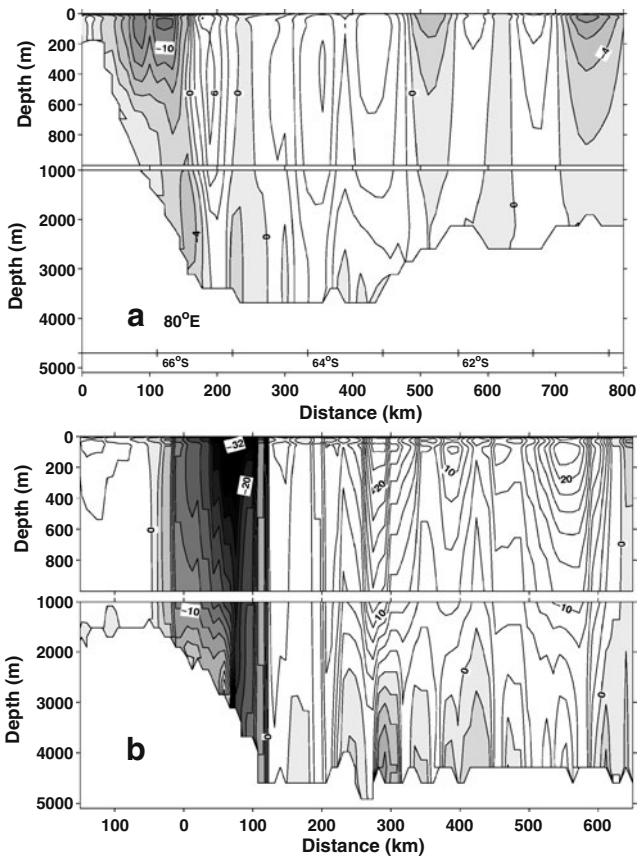

Fig. 7 Vertical sections of **a** the zonal velocity across 80° E and **b** the velocity perpendicular to the Kerguelen Plateau from the OFES. The location of the hydrographic section off the Kerguelen Plateau is shown in Fig. 1. Contour interval is 2 cm/s. Negative values are *shaded* for westward and southeastward flow in **a** and **b**, respectively

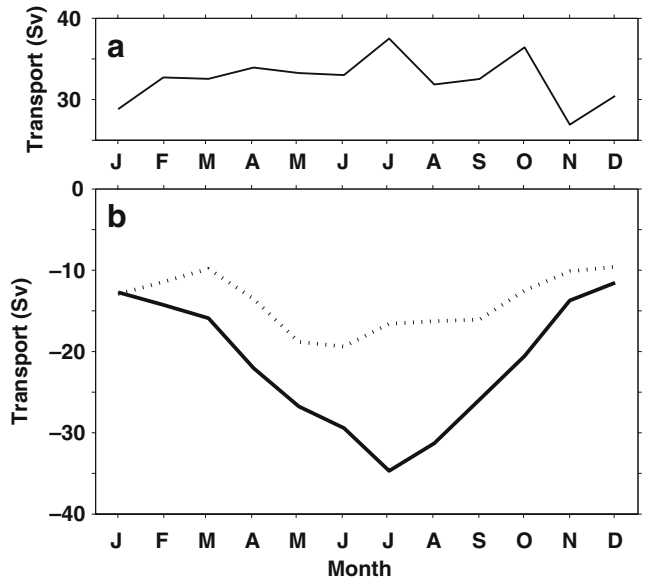

Fig. 9 Seasonal variations of the barotropic transport (Sverdrup). **a** The transport off the Kerguelen Plateau (*solid line*). **b** Both the transport between the center of the cyclonic gyre at 60.5° S, 100° E and the coast (*thick solid line*) and for eastern region between 64.5° S, 140° E and coast (*dotted line*). Negative values are westward, and the transport off the Kerguelen Plateau is the component perpendicular to the slope

decreases to 12 Sv in December/January. In the east, the seasonal variation has almost the same phase, but a weaker range from 19 Sv in June to 10 Sv in February. Off the Kerguelen Plateau, however, the transport shows no prominent seasonal variation.

4 Comparison of flow field with the Lagrangean trajectories

Trajectories of the icebergs and middepth floats are examined to reveal the realistic structure of the cyclonic circulation near the surface and in the deeper layer.

4.1 Surface drifts from the iceberg trajectories

Almost all icebergs drifted westward on the ASC along the slope (with the exception of iceberg UK-242; Fig. 1). Although the westward drift was dominant, eastward

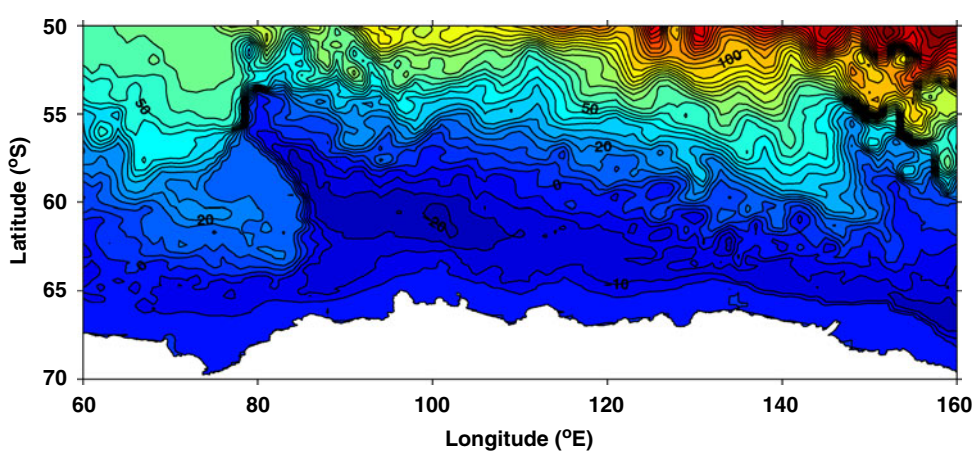

Fig. 8 Annual mean barotropic stream function around East Antarctica from the OFES. Contour interval is 5 Sv

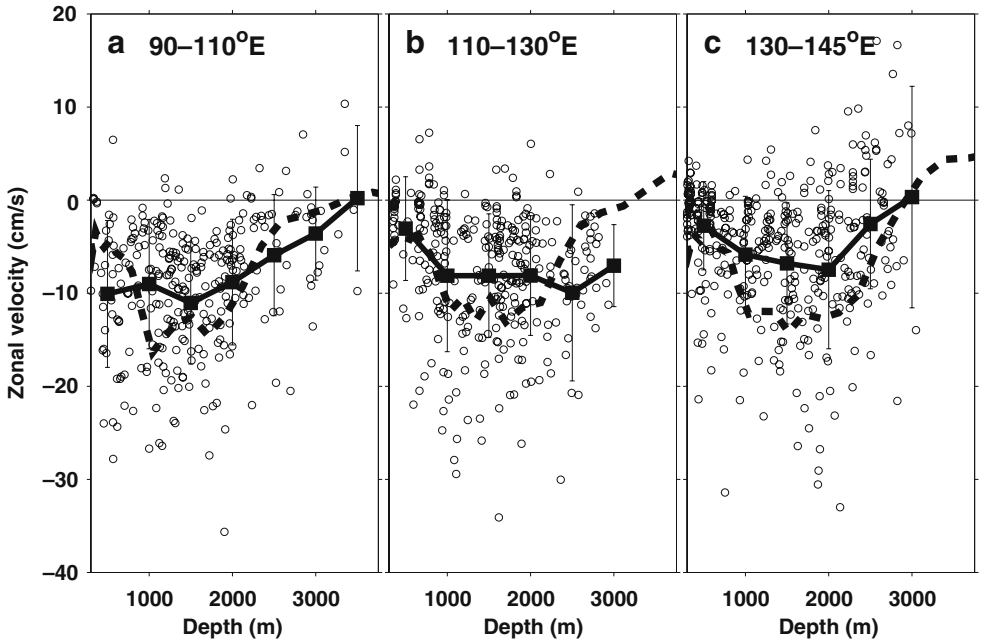

Fig. 10 Meridional variation in the zonal drift speed (centimeters per second) of icebergs with bottom depth over the continental slope in regions **a** 90–110° E, **b** 110–130° E, and **c** 130–145° E. Estimates made by individual drift trajectories are plotted by *circles* and their averages over the 500 m isobath are plotted with *squares*. *Broken lines* are those derived from the OFES output in Fig. 4

motion also occurred on the lower slope/rise in the eastern part. One example is the drift of iceberg C-19C; it moved from 148° E on 28 August 2004 to 151.6° E on 23 September 2004, and then it drifted westward to 148° E on 12 October 2004. Similar motion was detected in the region from 143° to 152° E. These icebergs drifted on the ACC on the lower slope/rise and maybe related to a series of cyclonic eddies (Aoki et al. 2007). However, all of the icebergs that moved eastward ultimately returned to the ASC and drifted westward afterward.

To examine the characteristics of the westward drift quantitatively, the region was divided into the three subregions, and the drift data were averaged spatially and temporally. Zonal drift speeds are plotted against the isobath (Fig. 10). Generally, the westward drift was dominant, but the drift speed had large variation in its range (up to 30 cm/s). The speed was generally larger

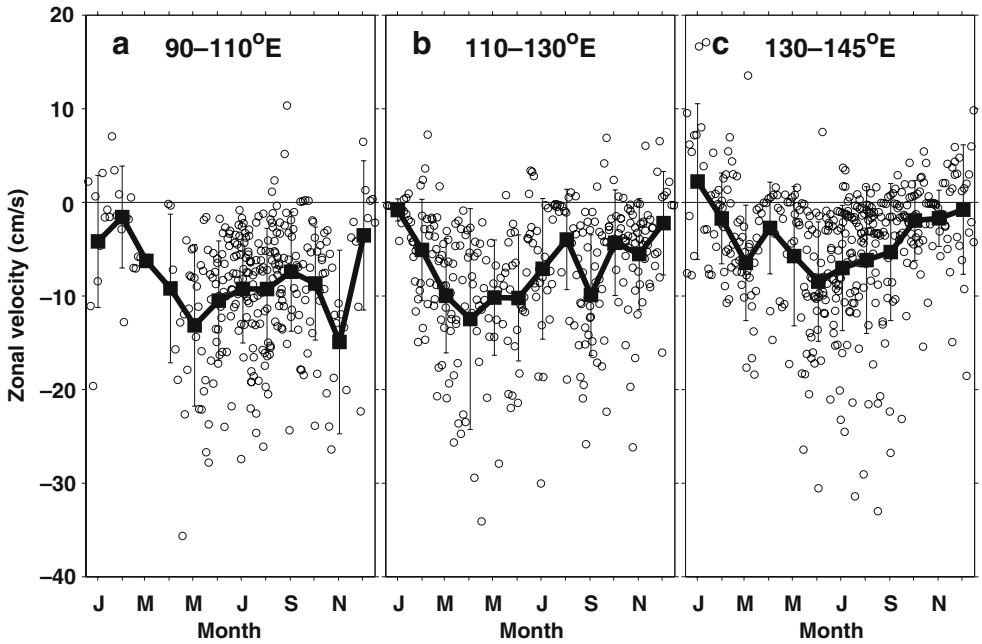

Fig. 11 Seasonal variation of the zonal component of the iceberg drift (centimeters per second) over the continental slope in regions **a** 90–110° E, **b** 110–130° E, and **c** 130–145° E. Estimates made by individual drift trajectories are plotted by *circles* and their monthly averages are plotted with *squares*

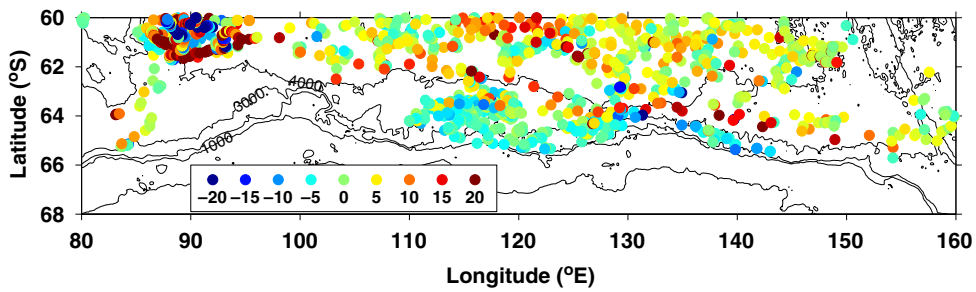

Fig. 12 Distribution of zonal drift speed (centimeters per second) from the Argo float trajectories. *Color of each dot* indicates the magnitude of the speed (negative westward)

on the upper slope, and the depth-averaged speed showed its maximum around the 1,000–2,000 m isobath. The maximum of the averaged speed was 10 cm/s in 90–110° and 110–130° E and 7 cm/s in 130–145° E. The maximum speed seemed to be slightly larger to the west but within the range of uncertainty. At the 3,000–3,500 m isobath, weaker drifts of 0–7 cm/s were found.

Seasonal variation of the drift speed was also prominent (Fig. 11). Maximum velocity was observed in April through June (10–12 cm/s in 90–110° and 110–130° E and 7–10 cm/s in 130–145° E) and minimum from December through January (0–4 cm/s in 90–110° and 110–130° E and −2–2 cm/s in 130–145° E).

The isolated iceberg UK-242, first detected offshore, showed eastward drift at around 120° E (Fig. 1). It drifted eastward from 119.3° E on 29 March 2008 to 120.8° E on 12 May 2008. The record was too short to completely trace the drift, but it was probably trapped by the eastward-flowing ACC.

After passing westward through the PET, seven icebergs (B-17B, C-6, C-20, E15, UK-95, UK-110, and UK-116) turned equatorward. This is not a trivial number, considering 11 icebergs drifted straight to the west of 60° E along the slope. The equatorward drift occurred at various longitudes from 77° to 89° E. Flow paths reflected the meanders and/or eddies in this region but generally showed a cyclonic recirculation pattern through the PET. Four of them then converged at around 63° S, 85° E over the southeast corner of the Kerguelen Plateau, while one drifted west of the plateau. The time taken from the slope to the plateau was from 80 (C-6) to 270 days (C-20), and hence, the averaged meridional drift speed was 1–5 cm/s.

After passing the southeast corner of the Kerguelen Plateau, three icebergs (C-6, C-20, and UK-95) drifted along the eastern flank of the Kerguelen Plateau. The drift speeds were 8–20 cm/s. Two icebergs drifted eastward at around 60° S, and one went equatorward at around 54° S along the Chun Spur. These drift paths were consistent with the locations of the major ACC fronts (Aoki et al. 2008).

4.2 Deep drifts from the middepth float trajectories

Middepth float (at 1,000 m) trajectories are examined to reveal the deep flow pattern of the gyre-scale circulation. For the offshore region, general eastward drift was obtained from the floats (Fig. 12). Latitudes of the eastward drift shifted poleward to the east; from 90° to 120° E, eastward drift was found along 60–61° S, and from 120° to 140° E, the eastward drift shifted poleward to 63–64° E. General westward drift was found on the slope from 110° to 145° E. In the region from 130°to 145° E, stronger westward drift was found on the lower slope with occasional eastward drifts due to the ACC. On the slope, the deep drift speed was averaged against the depth for the two subregions of 110–130° and 130–145° E (Fig. 13). For 110–130° E, a westward speed

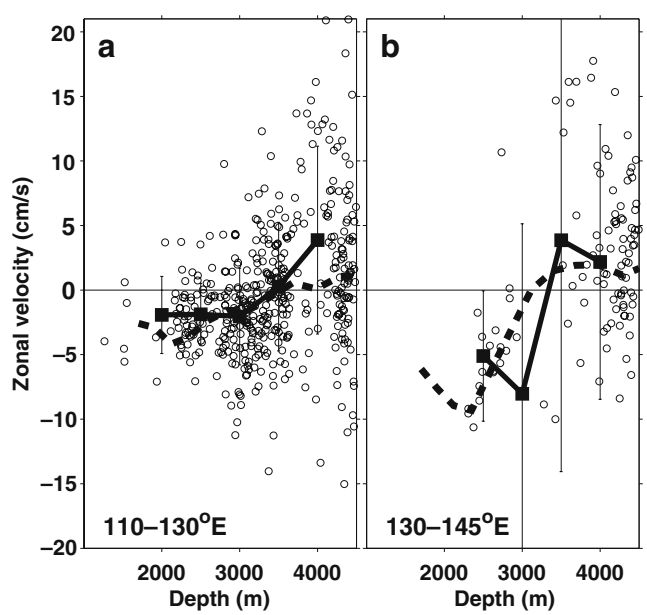

Fig. 13 Meridional variation in the zonal drift speed (centimeters per second) of middepth floats over the continental slope in regions **a** 110–130° E and **b** 130–145° E. Estimates made by individual floats are plotted by *circles* and their averages over the 500-m isobath are plotted with *squares*. *Broken lines* are those derived from the OFES output in Fig. 4

of 2 cm/s was detected at 2,000 m, but it was almost zero at 3,500 m. Over the 4,000 m isobath, the flow direction is reversed to eastward. For 130–145° E, the basic structure was the same, but a much larger drift of 7–10 cm/s was found over the 2,000–3,000 m isobath.

5 Discussion

The overall circulation system in OFES was consistent with that obtained from the drift patterns of icebergs and floats: a large-scale cyclonic circulation in the west, a series of cyclonic eddies in the east with the ACC closer, recirculation through the PET, and an equatorward excursion near the Kerguelen Plateau. Surface flow fields were consistent between the model and iceberg observations. In the southern boundary region, the maximum speed of the averaged ASC was around 15 cm/s for OFES and 10–12 cm/s from the iceberg data. These agreed well, given the large size (i.e., large spatial average) of the icebergs used in this study. Both fields indicated a stronger current in winter. In the western boundary region, the recirculation through the PET was also obtained and confirmed by iceberg drifts. Out of seven icebergs which drifted equatorward, no icebergs turned poleward to the coast again. This was consistent with the stream line extending through the PET rather than an isolated Prydz Bay Gyre whose presence was suggested by some previous studies. In the east, an eastward turn of icebergs near the coast between 143°and 152° E was also consistent with the ACC being located closer to the shore. The eastern closure of the large-scale structure in OFES, however, needs further investigation.

Deep drifts from the 1,000 m floats along the continental slope agreed well with the OFES flow field. Drift speed was derived as 2 cm/s around the 2,000 m depth between 110° and 130° E and 10 cm/s between 130° and 145° E, which were also consistent with the model. The intense deep current in the eastern region may not simply be wind-driven, and the intense ACC nearby might have some role on this structure. Seasonal variation of the deep drift was difficult to detect because their positions in winter could not be transmitted.

In addition to the Lagrangean tracers, the OFES flow field was compared quantitatively with the previous direct observations in and out of this basin. The intensity of the ASC was consistent with similar measurements obtained in the Weddell and Ross Seas (Jacobs 1991). Its seasonal variation was similar to that obtained in the Weddell–Enderby basin (Aoki 2003). The location of the ACC in the eastern region of OFES is consistent with the results from high-quality hydrographic surveys (e.g., Sokolov and Rintoul 2002).

Off the Adélie Land Coast at around 140° E, a year-long current measurements are available. Fukamachi et al. (2000) observed a mean current of 16.2 cm/s at 2,632 m and 5.3 cm/s at 1,075 m depth on the slope of 2,660 m. In OFES, a westward current of 9.5–12.0 cm/s was obtained at the 2,591 m depth at almost the same location. Hence, it showed reasonably good agreement, although the bottom intensified structure of the observations was not detected in the model. This is probably due to the fact that the bottom water formation (Rintoul 1998) was not properly reproduced in this model. Rintoul and Bullister (1999) and Bindoff et al. (2000) suggested a weakly westward flow near the bottom over the slope and rise in this region. This feature was revealed in OFES around the coast in this basin.

East of the Kerguelen Plateau, a current core is found on the slope and the maximum speed was 30 cm/s, and the reverse poleward flow exceeding 10 cm/s is simulated. This was similar to the results of one-time observations (Donohue et al. 1999; Aoki et al. 2008), but the estimates of the deeper current were slightly less than those observations. This may be due to a lack of spatial representation in the model and/or the short-term temporal variability that could not be excluded in a one-time hydrography.

In and near the PET, Kimura (2004) showed that an equatorward surface speed of about 5 cm/s was found between 63° and 61° S at 80–90° E. The OFES field shows an equatorward component whose speed is of similar magnitude. A hydrographic transect was occupied along 80° E in February 2006 during a research voyage "BROKE-West" (Fig. 14; Meijers et al. 2010). Velocity observations based on lowered acoustic Doppler current profiler (LADCP) revealed the strong westward ASC on about 1,500 m depth and change in flow direction at 64.5° S. Alternating flow patterns (eastward at 65.5° S and westward at 62° S) were obtained. These features were consistent with those revealed in the OFES. The bottom intensified westward current at 65° S was also consistent, although the observed magnitude of around 15 cm/s was larger than that of the model. A preliminary calculation based on the LADCP gave a cumulative westward transport of 30 Sv in the southern part (from the coast to 64.8° S) and eastward transport of 45 Sv in the northern part (from 64.8° to 62.7° S). The net transport through the PET was hence 15 Sv eastward. The observed westward

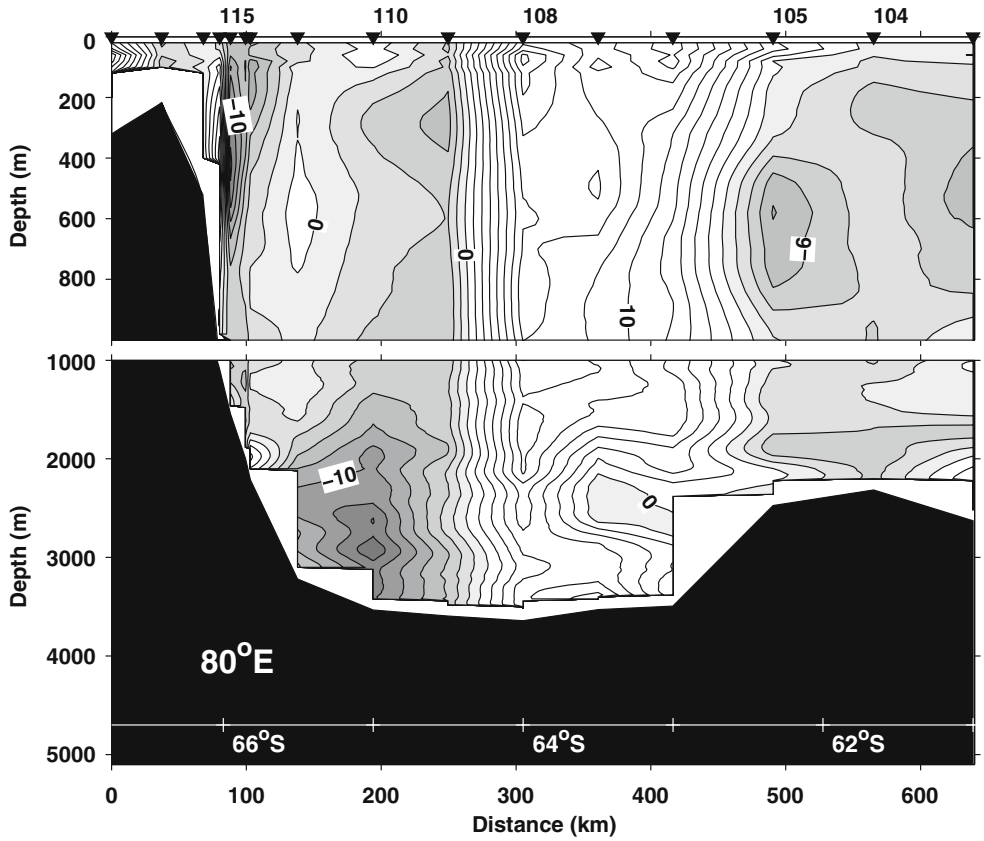

Fig. 14 Vertical section of the zonal velocity (centimeters per second) derived from lowered acoustic Doppler current profiler at CTD stations across 80° E in February 2006 during the "BROKE-West" research voyage. Station locations numbered and denoted by *black triangles*

and eastward transports were nearly double of the annual mean transports in the OFES, but the net eastward transport agreed well with 13 Sv in the OFES. For a transect along 85° E, Heywood et al. (1999) derived a westward transport of 45 Sv for the whole water column in the southern half of the PET. This is three times larger than the annual mean transport of the model and may be due to short-term temporal variability or a lack of resolution of the model near the bottom. They also observed an annual mean bottom current of 5.4–10.0 cm/s from year-long mooring observations. The bottom current in OFES was less than 3 cm/s for nearly the same locations and underestimated these observations. Hence, the bottom current was not well represented in this region of complex bathymetry. From a XCTD transect along 75° E of BROKE-West, a cyclonic eddy was located at 64.5° S off Prydz Bay (Williams et al. 2010). A similar mesoscale feature was revealed in the OFES, and hence, this can be persistent for a certain period of time.

Overall, the westward transport of the cyclonic circulation in the southern limb was around 22 Sv. This estimate is moderately large within the range (from 0 to 32 Sv) of the results of the IPCC models (Wang and Meredith 2008). It is in reasonably good agreement with 29 Sv of Bindoff et al. (2000) but significantly less than 76 Sv of McCartney and Donohue (2007). A strong wind stress curl in this region can lead to a large transport through a simple Sverdrup balance. The simple balance along 61° S integrated from 150° to 90° E, gave 93 (110) Sv from the NCEP (European Center for Medium-Range Weather Forecasts; Trenberth et al. 1989) wind stress field. However, in this basin, the bottom topography gradually deepens further to 60° S, and hence, the topographic beta effect (Thompson et al. 1986; Middleton and Cirano 2002) should be taken into account. A topographic Sverdrup transport was reduced to 29 (34) Sv considering the bottom slope of 0.0001, which was an average for the rise in this sector, and the steeper slope near the coast will further reduce the transport. Although the quantitative agreement with this simple model is limited due to the sensitivity on the bottom slope and selection of the latitude, the obtained value is much more realistic, and hence, the effect of the sloping bottom seems indispensable. Moreover, the processes that determine the upper

layer meridional density gradients should be taken into account for a quantitative explanation of the transport estimate (Wang and Meredith 2008).

6 Summary and conclusions

Spatial and temporal characteristics of a cyclonic circulation in the Australian–Antarctic basin were described using the output from OFES and validated with iceberg and mid-depth float trajectories. The ASC has a mean speed of around 15 cm/s and width of 100–150 km. The ASC has strong seasonal variability whose maximum speed occurred in winter. The ACC flows zonally in the western part and shifts poleward in the eastern part, after flowing along the eastern flank of the Kerguelen Plateau and recirculating immediately offshore. From these components, a cyclonic system consists of a large-scale (~1,000 km) gyre in the west and a series of eddies in the east. When integrating from the coast, the gyre had an annual mean transport of 22 Sv, half of which recirculated west of 110° E. About 5 Sv goes through the PET, turning equatorward north of Prydz Bay, and connects to the western boundary current off the Kerguelen Plateau.

This current system was examined quantitatively with Lagrangean observations of iceberg and middepth float trajectories. Iceberg drifts support the ASC structure with respect to isobath and its seasonal variation in the model, suggesting the extension of the western end of the cyclonic circulation through the PET. Middepth float trajectories also support a deep westward current of several centimeters per second, which was consistent with the model results.

The OFES results successfully reproduced the large-scale cyclonic circulation in this basin, which corresponds to the Weddell and Ross gyres. Further analysis of the dynamical balance of this circulation will lead to greater understanding of the difference in structure between the west and east regions. Moreover, it will help infer the cyclonic circulation in other regions under the circumpolar Antarctic Divergence between opposing wind regimes.

The strong seasonality of the transport in the southern limb suggests a dependence on the wind variability. This may explain the differences between the estimates derived from the one-time surveys like Aoki et al. (2008) and Heywood et al. (1999) and those of OFES with its temporal average. This implies the importance of long-term monitoring using moorings and repeated hydrographic surveys.

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